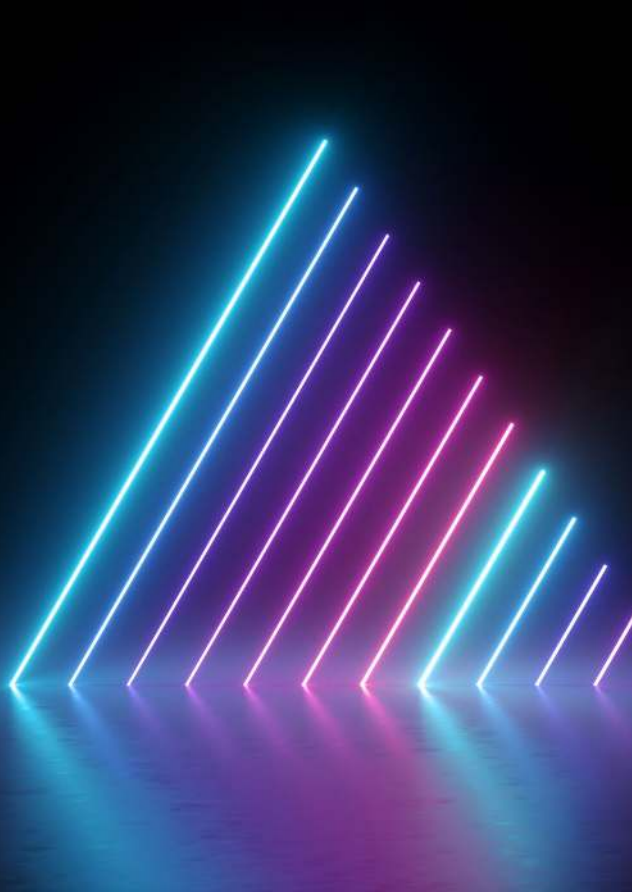




7.2 Eigenvalues and Eigenvectors



If A is a square matrix, the **characteristic polynomial** of A is defined by

$$p(\lambda) = \det(A - \lambda I).$$



It is not difficult to show (see Exercise 15) that p is an n th-degree polynomial and, consequently, has at most n distinct zeros, some of which might be complex. If λ is a zero of p , then, since $\det(A - \lambda I) = 0$, Theorem 6.17 on page 402 implies that the linear system defined by $(A - \lambda I)\mathbf{x} = \mathbf{0}$ has a solution with $\mathbf{x} \neq \mathbf{0}$. We wish to study the zeros of p and the nonzero solutions corresponding to these systems.

Definition 7.13

If p is the characteristic polynomial of the matrix A , the zeros of p are called **eigenvalues**, or characteristic values, of the matrix A . If λ is an eigenvalue of A and $\mathbf{x} \neq \mathbf{0}$ satisfies $(A - \lambda I)\mathbf{x} = \mathbf{0}$, then $\mathbf{x}$ is an **eigenvector**, or characteristic vector, of A corresponding to the eigenvalue λ . ■

en comes from the
tive meaning
synonymous in
the word

Each matrix has
or characteristic
corresponding
characteristic values

To determine the eigenvalues of a matrix, we can use the fact that

- λ is an eigenvalue of A if and only if $\det(A - \lambda I) = 0$.

Once an eigenvalue λ has been found, a corresponding eigenvector $\mathbf{x} \neq \mathbf{0}$ is determined by solving the system

- $(A - \lambda I)\mathbf{x} = \mathbf{0}$.

EXERCISE SET 7.2

1. Compute the eigenvalues and associated eigenvectors of the following matrices.

a. $\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$

b. $\begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$

c. $\begin{bmatrix} 0 & \frac{1}{2} \\ \frac{1}{2} & 0 \end{bmatrix}$

d. $\begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$

e. $\begin{bmatrix} -1 & 2 & 0 \\ 0 & 3 & 4 \\ 0 & 0 & 7 \end{bmatrix}$

f. $\begin{bmatrix} 2 & 1 & 1 \\ 2 & 3 & 2 \\ 1 & 1 & 2 \end{bmatrix}$

$$1) a) A = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$$

$$\underline{A - \lambda I} = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} = \begin{bmatrix} 2-\lambda & -1 \\ -1 & 2-\lambda \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 2-\lambda & -1 \\ -1 & 2-\lambda \end{bmatrix}$$

$$\begin{aligned} (2-\lambda)^2 &= \lambda^2 - 2(2\lambda) + 4 \\ \det(A - \lambda I) &= (2-\lambda)(2-\lambda) - (-1)(-1) \\ &= 4 - 4\lambda + \lambda^2 - 1 \\ &= \lambda^2 - 4\lambda + 3 \\ \lambda^2 - 4\lambda + 3 &= 0 \\ \lambda &= 1, 3 \end{aligned}$$

For $\lambda=1$

$$\begin{bmatrix} 2-1 & -1 \\ -1 & 2-1 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \xrightarrow{R_1+R_2} \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}$$

$1-1=0 \quad -1+1=0$

$$\begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$x_1 - x_2 = 0$$

$$x_1 = x_2$$

$$\begin{bmatrix} 1 \\ 1 \end{bmatrix} = (1, 1)^t$$

For $\lambda=3$

$$-(-1) - 1 = 1 - 1 = 0$$

$$\begin{bmatrix} 2-3 & -1 \\ -1 & 2-3 \end{bmatrix} = \begin{bmatrix} -1 & -1 \\ -1 & -1 \end{bmatrix} \xrightarrow{-R_1+R_2} \begin{bmatrix} -1 & -1 \\ 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} -1 & -1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$-x_1 - x_2 = 0$$

$$x_1 = -x_2$$

$$\begin{bmatrix} -1 \\ 1 \end{bmatrix} = (-1, 1)^t$$

$\lambda=1$ eigenvector is $(1, 1)^t$

$\lambda=3$ eigenvector is $(-1, 1)^t$

$$1) \neq) A = \begin{bmatrix} 2-\lambda & 1 & 1 \\ 2 & 3-\lambda & 2 \\ 1 & 1 & 2-\lambda \end{bmatrix} \quad A - \lambda I = \begin{bmatrix} 2 & 1 & 1 \\ 2 & 3 & 2 \\ 1 & 1 & 2 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 1 \\ 2 & 3 & 2 \\ 1 & 1 & 2 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix} = \begin{bmatrix} 2-\lambda & 1 & 1 \\ 2 & 3-\lambda & 2 \\ 1 & 1 & 2-\lambda \end{bmatrix}$$

$$\begin{vmatrix} 2-\lambda & 1 & 1 & 2-\lambda & 1 \\ 2 & 3-\lambda & 2 & 2 & 3-\lambda \\ 1 & 1 & 2-\lambda & 1 & 1 \end{vmatrix} \begin{matrix} - & - & - & + & + \end{matrix}$$

$$\det(A - \lambda I) = (2-\lambda)(3-\lambda)(2-\lambda) + (1)(2)(1) + (1)(2)(1) - (1)(3-\lambda)(1) - (2-\lambda)(2)(1) - (1)(2)(2-\lambda)$$

$$= (2-\lambda)^2(3-\lambda) + 2 + 2 - (3-\lambda) - (4-2\lambda) - (4-2\lambda)$$

$$= (4-4\lambda+\lambda^2)(3-\lambda) + 4 - 3 + \lambda - 4 + 2\lambda - 4 + 2\lambda$$

$$= (12-12\lambda+3\lambda^2-4\lambda+4\lambda^2-\lambda^3) - 7 + 5\lambda$$

$$= -\lambda^3 + 7\lambda^2 - 11\lambda + 5$$

$$-\lambda^3 + 7\lambda^2 - 11\lambda + 5 = 0$$

$$\lambda = 1, 5$$

For $\lambda=1$

$$A - \lambda I = \begin{bmatrix} 2-1 & 1 & 1 \\ 2 & 3-1 & 2 \\ 1 & 1 & 2-1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 1 & 1 & 1 \end{bmatrix} \xrightarrow{\substack{-2R_1 + R_2 \\ -R_1 + R_3}} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$x_1 + x_2 + x_3 = 0$$

$$x_1 = -x_2 - x_3$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -x_2 - x_3 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -x_2 \\ x_2 \\ 0 \end{bmatrix} + \begin{bmatrix} -x_3 \\ 0 \\ x_3 \end{bmatrix}$$

$$= x_2 \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

eigen vectors

$\lambda=1$ eigen vectors are $(-1, 1, 0)^t, (-1, 0, 1)^t$

$\lambda=5$ eigen vector is $(1, 2, 1)^t$

For $\lambda=5$

$$A - \lambda I = \begin{bmatrix} 2-5 & 1 & 1 \\ 2 & 3-5 & 2 \\ 1 & 1 & 2-5 \end{bmatrix} = \begin{bmatrix} -3 & 1 & 1 \\ 2 & -2 & 2 \\ 1 & 1 & -3 \end{bmatrix} \xrightarrow{\substack{R_1 \leftrightarrow R_3 \\ -2R_1 + R_2 \\ 3R_1 + R_3}} \begin{bmatrix} 1 & 1 & -3 \\ 2 & -2 & 2 \\ -3 & 1 & 1 \end{bmatrix} \xrightarrow{\substack{-2R_1 + R_2 \\ 3R_1 + R_3}} \begin{bmatrix} 1 & 1 & -3 \\ 0 & -4 & 8 \\ 0 & 4 & -8 \end{bmatrix} \xrightarrow{-R_2 + R_3} \begin{bmatrix} 1 & 1 & -3 \\ 0 & -4 & 8 \\ 0 & 0 & 0 \end{bmatrix}$$

$-2R_1 \leftrightarrow R_2$
 $3R_1 + R_3$

$$\begin{bmatrix} 1 & 1 & -3 \\ 0 & -4 & 8 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$-4x_2 + 8x_3 = 0$$

$$\frac{4x_2}{4} = \frac{8x_3}{4}$$

$$x_2 = 2x_3$$

$$x_1 + x_2 - 3x_3 = 0$$

$$x_1 + 2x_3 - 3x_3 = 0$$

$$x_1 - x_3 = 0$$

$$x_1 = x_3$$

$$\text{eigen vector } \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} = (1, 2, 1)^t$$

7.2

①① $A = \begin{bmatrix} 2 & 1 & k \\ 2 & 3 & 2 \\ 1 & 1 & 2 \end{bmatrix}$

$$A - \lambda I = \begin{bmatrix} 2 & 1 & 1 \\ 2 & 3 & 2 \\ 1 & 1 & 2 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 2-\lambda & 1 & 1 \\ 2 & 3-\lambda & 2 \\ 1 & 1 & 2-\lambda \end{bmatrix}$$

$$\det(A - \lambda I) = \begin{vmatrix} 2-\lambda & 1 & 1 & | & 2-\lambda & -1 \\ 2 & 3-\lambda & 2 & | & 2 & 3-\lambda \\ 1 & 1 & 2-\lambda & | & 1 & -1 \end{vmatrix}$$

$$= (2-\lambda)(3-\lambda)(2-\lambda) + 2 + 2 - 2(2-\lambda) - 2(2-\lambda) - (3-\lambda)$$

$$= (2-\lambda)^2(3-\lambda) + 4 - 4 + 2\lambda - 4 + 2\lambda - 3 + \lambda$$

$$= (2-\lambda)^2(3-\lambda) + 5\lambda - 7$$

$$= (4 - 4\lambda + \lambda^2)(3-\lambda) + 5\lambda - 7$$

$$= 12 - 12\lambda + 3\lambda^2 - 4\lambda + 4\lambda^2 - \lambda^3 + 5\lambda - 7$$

$$= -\lambda^3 + 7\lambda^2 - 11\lambda + 5$$

$$\begin{matrix} 1 & 0 & 0 & 2 & x_1 = 2 \\ 0 & 1 & 0 & 3 & x_2 = 3 \\ 0 & 0 & 1 & 4 & x_3 = 4 \end{matrix}$$

$\det(A - \lambda I) = 0$
 $-\lambda^3 + 7\lambda^2 - 11\lambda + 5 = 0$
 $\lambda = 1, 5$ eigen values.

For $\lambda = 1$
 $A - \lambda I = \begin{bmatrix} 2 & -1 & 1 & 1 \\ 2 & 3 & -1 & 2 \\ 1 & 1 & 2 & -1 \end{bmatrix}$
 $\xrightarrow{R_1, R_2, R_3} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 2 & 2 & 2 \\ 1 & 1 & 1 \end{bmatrix} \xrightarrow{R_2 - 2R_1, R_3 - R_1} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$
 $2 - 2(1) = 0$
 $2 - 2(1) = 0$
 $2 - 2(1) = 0$

$X_1 + X_2 + X_3 = 0 \Rightarrow X_1 = -X_2 - X_3$
 $X_2 = 0$
 $X_1 + X_3 = 0$
 $X_1 = -X_3$
 eigen vector $\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$

$\lambda = 5$
 $A - \lambda I = \begin{bmatrix} -3 & -1 & 1 & 1 \\ 2 & -2 & 2 \\ 1 & 1 & -3 \end{bmatrix}$
 $\xrightarrow{-2R_1 + R_2, -R_1 + R_3} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$
 $X_1 + X_2 + X_3 = 0$
 taking $X_3 = 0$
 $X_1 + X_2 = 0$
 $X_1 = -X_2$
 eigen vector $\begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$

$\begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix} = \begin{bmatrix} -X_2 - X_3 \\ X_2 \\ X_3 \end{bmatrix} = \begin{bmatrix} -X_2 \\ X_2 \\ 0 \end{bmatrix} + \begin{bmatrix} -X_3 \\ 0 \\ X_3 \end{bmatrix} = X_2 \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + X_3 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$
 eigen vectors

$$\begin{array}{cc}
 X_1 + X_2 + X_3 = 0 & \\
 \downarrow & \downarrow \\
 \text{take } X_3 = 0 & \text{take } X_2 = 0 \\
 X_1 + X_2 = 0 & X_1 + X_3 = 0 \\
 X_1 = -X_2 & X_1 = -X_3 \\
 \text{1st eigen vector } \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} & \text{2nd eigen vector } \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}
 \end{array}$$

$$R_1 \leftrightarrow R_3 \rightarrow \begin{bmatrix} 1 & 1 & -3 \\ 2 & -2 & 2 \\ -3 & 1 & 1 \end{bmatrix}$$

$$\begin{matrix} -2R_1 + R_2 \\ 3R_1 + R_3 \end{matrix} \rightarrow \begin{bmatrix} 1 & 1 & -3 \\ 0 & -4 & 8 \\ 0 & 4 & -8 \end{bmatrix}$$

$$R_2 + R_3 \rightarrow \begin{bmatrix} 1 & 1 & -3 \\ 0 & -4 & 8 \\ 0 & 0 & 0 \end{bmatrix}$$

$$-4x_2 + 8x_3 = 0 \Rightarrow \frac{4x_2}{4} = \frac{8x_3}{4}$$

$$x_2 = 2x_3$$

$$x_1 + x_2 - 3x_3 = 0$$

$$x_1 + 2x_3 - 3x_3 = 0$$

$$x_1 - x_3 = 0$$

$$x_1 = x_3$$

$$\text{eigen vector} \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} = (1, 2, 1)^T$$

$\lambda = 1$ eigenvectors are

$$(1, -1, 0)^T, (1, 0, -1)^T \leftarrow \text{transpose}$$

$\lambda = 5$ eigenvector is $(1, 2, 1)^T$